

Simulation Of Preemptive Priority Queuing Model

M/M/1

P#	Arrival Time	Service Time	Priority	Start	End	Turn Around	Wait Time	Response Time
1	0	19	3	0	70	70	51	0
2	4	18	3	70	88	84	66	66
3	7	12	2	7	30	23	11	0
4	9	17	3	88	105	96	79	79
5	15	5	1	15	20	5	0	0
6	18	6	1	20	26	8	2	2
7	27	13	2	30	43	16	3	3
8	30	15	2	43	58	28	13	13
		105			330	225	163	

GENERAL:

Avg Turn Around = $330/8 = 41.25$

Avg Wait Time = $225/8 = 28.125$

Avg Response Time = $163/8 = 20.375$

Avg Service Time = $105/8 = 13.125$

Priority Wise:

Avg Turn Around:

1 Priority $\Rightarrow 2 = 8+5/2 = 6.5$

2 Priority $\Rightarrow 3 = 23+16+28/3 = 22.33$

3 Priority $\Rightarrow 3 = 70+84+96/3 = 83.33$

Avg Wait Time:

1 Priority $\Rightarrow 2 = 0+2/2 = 1$

2 Priority $\Rightarrow 3 = 11+13+3/3 = 9$

3 Priority $\Rightarrow 3 = 51+66+79/3 = 65.33$

Avg Response Time:

1 Priority $\Rightarrow 2 = 0+2/2 = 1$

2 Priority $\Rightarrow 3 = 3+13+0/3 = 5.33$

3 Priority $\Rightarrow 3 = 0+66+79/3 = 48.33$

Avg Service Time:

1 Priority $\Rightarrow 2 = 5+6/2 = 5.5$

2 Priority $\Rightarrow 3 = 12+13+15/3 = 13.33$

3 Priority $\Rightarrow 3 = 19+18+17/3 = 18$

Probability of waiting patients = $7/8 = 0.8 = 80\%$

M/M/2

P#	Arrival Time	Service Time	Priority	Server	Start	End	Turn Around	Wait Time	Response Time
1	0	19	3	A	0	47	47	28	0
2	4	18	3	B	4	28	24	6	0
3	7	12	2	A	7	24	17	5	0
4	9	17	3	B	45	62	53	36	36
5	15	5	1	A	15	20	5	0	0
6	18	6	1	B	18	24	6	0	0
7	27	13	2	A	27	40	13	0	0
8	30	15	2	B	30	45	15	0	0
		105					180	75	36

GENERAL:

Avg Turn Around = $180/8 = 22.5$

Avg Wait Time = $75/8 = 9.375$

Avg Response Time = $36/8 = 4.5$

Avg Service Time = $105/8 = 13.125$

Priority Wise:

Avg Turn Around:

1 Priority => $2 = 5+6/2 = 5.5$

2 Priority => $3 = 17+13+15/3 = 15$

3 Priority => $3 = 47+24+53/3 = 41.333$

Avg Wait Time:

1 Priority => $2 = 0+0/2 = 0$

2 Priority => $3 = 5+0+0/3 = 1.666$

3 Priority => $3 = 28+6+36/3 = 23.33$

Avg Response Time:

1 Priority => $2 = 0+0/2 = 0$

2 Priority => $3 = 0+0+0/3 = 0$

3 Priority => $3 = 36+0+0/3 = 12$

Avg Service Time:

1 Priority => $2 = 5+6/2 = 5.5$

2 Priority => $3 = 12+13+15/3 = 13.33$

3 Priority => $3 = 19+18+17/3 = 18$

Probability of waiting patients = $4/8 = 0.5 = 50\%$

FORMULA:

• Turn Around = End – Arrival Time.

• Wait Time = Turn Around – Service Time.

• Response Time = Start – Arrival Time.

